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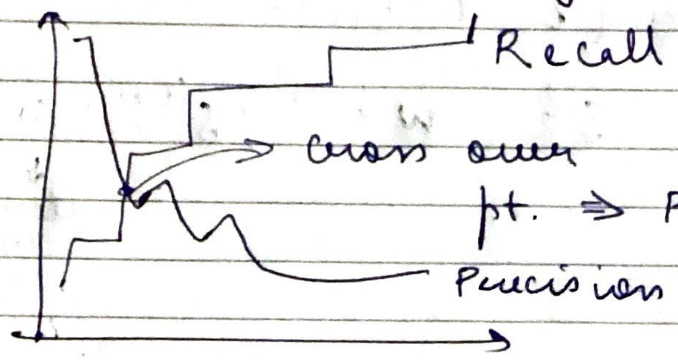
3rd Feb 2026 | CS6370

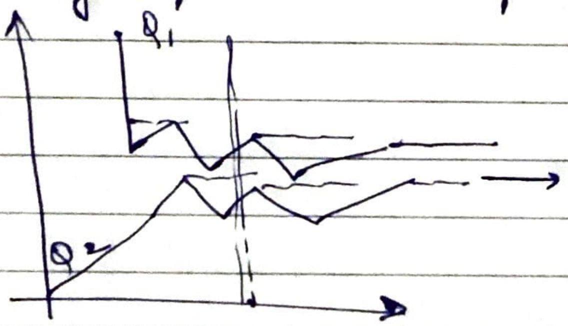
(IR Evaluaⁿ)

- When there is class imbalance, the accuracy as a metric does not make much sense.
- Recall / Precision

It should be at the same rank. Give higher impt. to PR at higher ranks.



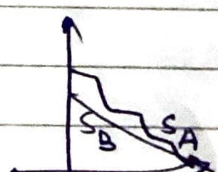
- For each query, plot the PR pts.



convert them into monotone property

now we can move to the averaging method.

It is done at 11-pts.



which sys. is better if you want P greater or R better?

if all the pts. are within then, it is fine. if they are intersecting, then, we can't judge them just by looking. We have to extrapolate & then check.

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- if the graphs are non-overlapping, then it is easier to claim which sys. is better.

- MAP (Mean Avg. Precision)

→ over ranks

over query

✓ ✓ – – ✓ – –

q_1

1 0.67 0.5

avg. precision q_1 = $\frac{1 + 0.67 + 0.5}{3} = a_1$

total no. of relevant docs

✓ – ✓ – – –

0.5 0.4

avg. precision q_2 = $\frac{0.5 + 0.4}{2} = a_2$

R → Relevant

N → Non-"

R N R N R

R R R N N ✓

this is better because relevant docs. are present at the top of the rank itself

$$\boxed{MAP = \frac{a_1 + a_2}{2}}$$

MAP is simply the avg. of AP (average precision) scores across multiple classes.

Precision → how correct are my predictions

AP → how good is my ranking for 1 class.

mAP → how good is my model overall across all classes

→ so far binary relevance info.

- nDCG (normalized Discounted Cumulative Gain)

For certain query, ⇒

i	rel_i	$\log_2(i+1)$	$rel_i / \log_2(i+1)$
1	3	1	3
2	2	1.585	1.262
3	3	2	1.5

$$DCG_6 = \sum_{i=1}^{6} \frac{rel_i}{\log_2(i+1)} = 3 + 1.262 + 1.5 = d_1$$

this can be any no., they are not in any particular range.

Ideal ordering : 3, 3, 2, 2, 1, 0

based on → Ideal DCG_6

$IDCG_6 = 7.141$

$$\boxed{nDCG_6 = \frac{d_1}{7.141}}$$

✓ Fill the feedback form.

- nDCG → measures how good a ranked list of results is.
  → it answers how well did the system rank the most relevant items near the top.

Components

1. gain (relevance score) → each result has a relevance score
2. DCG → higher wts to results at the top of rank